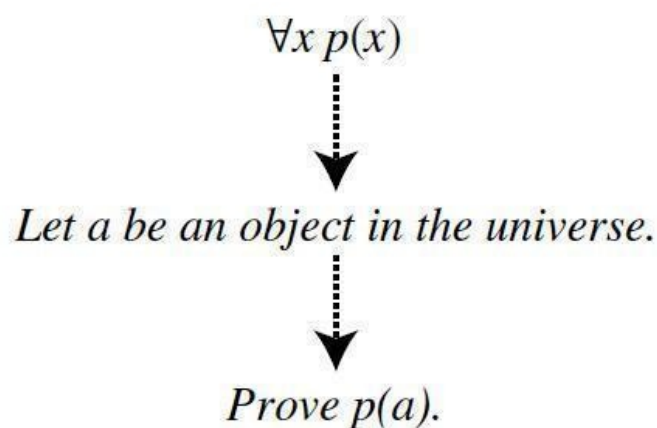


Universal Proofs

To prove $\forall x p(x)$ from a given set of premises, we show that every object satisfies $p(x)$ assuming those premises. Since the proofs are mathematical, we can restrict the objects to a particular **universe**. A proposition of the form $\forall x p(x)$ is then interpreted to mean that $p(x)$ holds *for all x from a given universe*.



These types of proofs are called **universal proofs**.

Note:

Definition:

For all integers a and b , a **divides** b (written as $a \mid b$) if $a \neq 0$ and there exists an integer k such that $b = ak$.

Note 1. $a \mid b \iff \exists k \in \mathbb{Z}, b = ak$
 2. $a \nmid b \iff \neg(\exists k \in \mathbb{Z}, b = ak) \equiv \forall k \in \mathbb{Z}, b \neq ak$

Definition:

An integer n is **even** if $2 \mid n$, and n is **odd** if $2 \nmid (n - 1)$.

Example 1:

Prove that for all even integer n , n^2 is even.

Proof: Let a be an even integer.
 Then, $a = 2k$ for some $k \in \mathbb{Z}$ (By the definition of even)
 $\therefore a^2 = 2(2k^2)$
 $\therefore a^2$ is even (By the definition of even)
 $\therefore$ For any even integer n , n^2 is even.

Example 2:

Prove that for all real number x , $x^2 - 12x + 37 > 0$

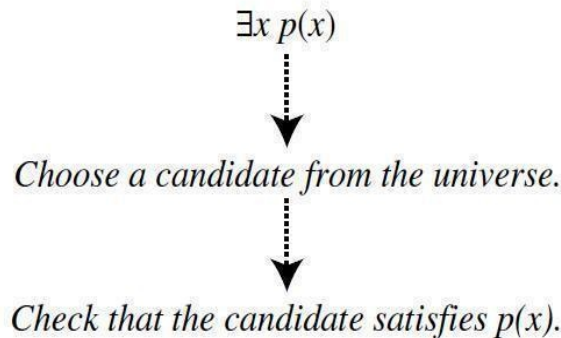
Prove that $\forall x \in \mathbb{R}, x^2 - 12x + 37 > 0$. Proof: Let $a \in \mathbb{R}$.

Existential Proofs

$$\begin{aligned} a^2 - 12a + 37 &= (a-6)^2 - 36 + 37 \\ &= (a-6)^2 + 1 > 1 > 0 \\ \therefore a^2 - 12a + 37 > 0 \quad \forall a \in \mathbb{R} \end{aligned}$$

Suppose that we want to write a proof for $\exists x p(x)$. This means that we must show that there exists at least one object of the universe that satisfies the formula $p(x)$.

It will be our job to find that object. To do this directly, we pick an object that we think will satisfy $p(x)$. This object is called a **candidate**. We then check that it does satisfy $p(x)$. This type of a proof is called a **direct existential proof**, and its structure is illustrated as follows:



Example 1:

Prove that there exists an integer x such that $x^2 + 2x - 3 = 0$.

Proof: Choose $x=1$. Then $x \in \mathbb{Z}$ and $1^2 + 2 \cdot 1 - 3 = 0$.
 $\therefore \exists x \in \mathbb{Z}, x^2 + 2x - 3 = 0$.

or Proof: Put $x=-3$.
Then $x \in \mathbb{Z}$ and $(-3)^2 + 2(-3) - 3 = 9 - 6 - 3 = 0$.
 $\therefore \exists x \in \mathbb{Z}, x^2 + 2x - 3 = 0$.

Example 2:

Prove that there exists real number x such that $x^{2026} = x$.

Proof: Choose $x=0$.
Then, $x \in \mathbb{R}$ and $0^{2026} = 0$.
 $\therefore \exists x \in \mathbb{R}, x^{2026} = x$.

Existence and Uniqueness

There will be times when we want to show that there exists exactly one object that satisfies a given predicate $p(x)$. In other words, there exists a **unique** object that satisfies $p(x)$. This is a two-step process.

- **Existence:** Show that there is at least one object that satisfies $p(x)$.
- **Uniqueness:** Show that there is at most one object that satisfies $p(x)$. This is usually done by assuming both a and b satisfy $p(x)$ and then proving $a = b$.

Example 1:

There exists an even prime number.

Ans: choose $p=2$.
Then, p is even and p is prime.

Example 2:

There exists an integer that can be expressed as the sum of two perfect cubes in two different ways.

Proof. Consider the number 1729. Note that $1^3 + 12^3 = 1729$ and $9^3 + 10^3 = 1729$. Thus the number 1729 can be expressed as the sum of two perfect cubes in two different ways. ■

Srinivasa Ramanujan (1887–1920) was a famous Indian mathematician. One day, G. H. Hardy visited him in hospital in a taxi numbered 1729. Hardy said it was a boring number, but Ramanujan quickly replied that 1729 is special because it is the *smallest number that can be written as the sum of two cubes in two different ways*. Since then, 1729 is called the Ramanujan number or the Hardy–Ramanujan number.

$$\begin{aligned} 1729 &= 12^3 + 1^3 = 10^3 + 9^3 \\ 12^3 + 1^3 &= 1728 + 1 = 1729 \\ 10^3 + 9^3 &= 1000 + 729 = 1729 \end{aligned}$$

Example 3:

Prove that $2x + 1 = 5$ exists real solution, and it is unique.

Proof: choose $x=2$.

Then $x \in \mathbb{R}$ and $2 \times 2 + 1 = 4 + 1 = 5$

$\therefore \exists x \in \mathbb{R}, 2x + 1 = 5$.

Suppose $\exists a, b \in \mathbb{R}$ such that $2a + 1 = 5$ and $2b + 1 = 5$

Then, $2a + 1 = 2b + 1$

$\therefore 2a = 2b$

$\therefore a = b$

$\therefore \exists! x \in \mathbb{R}, 2x + 1 = 5$

Note:

The phrase *exactly one* occurs so often in mathematics that there is a special notation for it.

We will write $\exists! x, P(x)$ to represent the statement

“There is exactly one value of x such that $P(x)$ is true.”

It is sometimes also read “There is a unique x such that $P(x)$.”

For example, the $2x + 1 = 5$ has a unique real solution.

i.e. $\exists! x \in \mathbb{R}, 2x + 1 = 5$.

Direct proofs

This gives a fundamental outline for proving statements of the form $P \Rightarrow Q$. Begin by assuming that P is true (remember, we don't need to worry about P being false) and show this forces Q to be true.

Outline for Direct Proof

Proposition If P , then Q .

Proof. Suppose P .

$\vdots$

Therefore Q . ■

So the setup for direct proof is remarkably simple. The first line of the proof is the sentence "*Suppose P .*" The last line is the sentence "*Therefore Q .*" Between the first and last line we use logic, definitions and standard math facts to transform the statement P to the statement Q . It is common to use the word "*Proof*" to indicate the beginning of a proof, and the symbol ■ to indicate the end.

Example 1: Let x be an integer. Prove that if x is odd, then $x + 1$ is even.

Example 2: Suppose a , b , and c are integers.

Prove that if a divides b and b divides c , then a divides c .

Example 3: Let a and b be positive real numbers.

Prove that if $a < b$, then $b^2 - a^2 > 0$.

Example 4: If x is an even integer, then $x^2 - 6x + 5$ is odd.

Direct proof of " $\forall x \in D, P(x) \Rightarrow Q(x)$ ":

Let $x \in D$ -----(i)

Suppose that $P(x)$. -----(ii)

$\left. \begin{array}{l} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \right\} \begin{array}{l} \text{Definitions,} \\ \text{Axioms,} \\ \text{Logical equivalences,} \\ \text{Rules of inference} \end{array}$

Therefore, $Q(x)$. -----(iii)

By (i), (ii), (iii) we get $\forall x \in D, P(x) \Rightarrow Q(x)$. ■

Example 1: Prove that for all $n \in \mathbb{N}$, if n is even, then n^2 is even.

i.e. Prove that " $\forall n \in \mathbb{N}, n \text{ is even} \Rightarrow n^2 \text{ is even}$."

Example 2: For all $n \in \mathbb{Z}$, if $3n + 5$ is even, then $3n$ is odd.

i.e. " $\forall n \in \mathbb{Z}, 3n+5 \text{ is even} \Rightarrow 3n \text{ is odd}$ "

Example 3: For all $n \in \mathbb{Z}$, if $3n + 5$ is even, then n is odd. (Ex)

Example 4: For all $x \in \mathbb{R}, x > 1 \Rightarrow x^2 > x$.

Proof: Suppose $a|b$ and $b|c$ (i)
Then, $b = ak_1$ and
 $c = bk_2$ for some
 $k_1, k_2 \in \mathbb{Z}$

$\therefore c = (ak_1)k_2$
 $\therefore c = ak$, where $k = k_1k_2$
 $\therefore a|c$ (ii)

By (i), (ii) $a|b$ and $b|c \Rightarrow a|c$ ■

Proof: Suppose x is an even integer (i)

Then, $2|x$

$\therefore \exists k \in \mathbb{Z}, x = 2k$

$\therefore x^2 - 6x + 5 = 4k^2 - 12k + 5$
 $= 2(2k^2 - 6k + 2) + 1$

Hence, $2|(x^2 - 6x + 5) - 1$

$\therefore x^2 - 6x + 5$ is odd (ii)

By (i), (ii) if x is even, then $x^2 - 6x + 5$ is odd ■

Proof: Let $n \in \mathbb{Z}$ (i)

Suppose $3n+5$ is even (ii)

$2|3n+5$ (By the definition of even)

$\therefore 3n+5 = 2k$, where $k \in \mathbb{Z}$

$3n-1 = 2k-6$, where $k \in \mathbb{Z}$
 $= 2(k-3)$
 $= 2n$, where $n = k-3 \in \mathbb{Z}$

$\therefore 2|3n-1$ (iii)

$\therefore 3n$ is odd (By defn)

$\forall n \in \mathbb{Z}, 3n+5 \text{ is even} \Rightarrow 3n \text{ is odd}$ ■

Proof: Let $x \in \mathbb{R}$ — (i)
Suppose $x > 1$ — (ii)

(Ex)

$\therefore x^2 > x$ — (iii)

By (i), (ii) and (iii) we have $\forall x \in \mathbb{R}, x > 1 \Rightarrow x^2 > x$ $\square$

End of Dg